Can Small Models Learn to Think? Reasoning Distillation vs. Reinforcement Learning in a 1.7B Parameter Model

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Abstract

Large reasoning models such as DeepSeek-R1 have demonstrated that explicit chain-of-thought reasoning can emerge through reinforcement learning and subsequently transfer to smaller models via distillation. However, the relative contributions of these two training signals—supervised imitation of reasoning traces versus reinforcement learning with verifiable rewards—remain poorly understood, particularly at the sub-2B parameter scale where model capacity is most constrained. We present a controlled study comparing five training conditions applied to Qwen3-1.7B: (1) supervised fine-tuning on answer-only data (baseline, 67.0%), (2) SFT on DeepSeek-R1 reasoning traces (distillation, 80.8%), (3) Group Relative Policy Optimization from the base model (pure RL, 77.8%), (4) distillation followed by GRPO (combined, 66.7%), and (5) SFT on RL-verified traces from Open-R1 (re-distillation, 86.0%). All conditions share the same base model, LoRA configuration, and evaluation protocol, isolating the training signal as the sole variable. We find that distillation alone outperforms direct RL, that adding GRPO after distillation *degrades* performance by 14 points through catastrophic forgetting, and that re-distillation from RL-verified traces achieves the best results. The path to strong reasoning in small models runs through data quality, not RL optimization: one thousand verified traces and a single SFT run ($2 on Modal) outperform weeks of GRPO engineering and $40+ of H100 compute.

1 Introduction

The release of DeepSeek-R1 [5] marked a turning point for reasoning in language models. By applying Group Relative Policy Optimization (GRPO) to a base model without any supervised reasoning data, DeepSeek demonstrated that explicit chain-of-thought reasoning can emerge from reinforcement learning alone—and that the resulting capabilities transfer to models as small as 1.5B parameters through supervised distillation of reasoning traces. Shortly after, Muennighoff et al. [16] showed that fine-tuning on as few as 1,000 curated DeepSeek-R1 traces, combined with test-time budget forcing, enables a 32B model to match OpenAI's o1-preview on competition mathematics. These results have triggered a wave of distillation efforts, with practitioners fine-tuning small models on publicly available reasoning traces under the assumption that distillation is sufficient for transferring reasoning capabilities.

Yet this assumption remains largely untested in controlled settings. Distillation is cheap and scalable, but several lines of evidence suggest it has hard limits. Li et al. [11] demonstrate a *learnability gap*: small models struggle to absorb the full reasoning structure of traces generated by much larger models, often learning surface patterns rather than transferable problem-solving strategies. Conversely, Dang and Ngo [4] show that reinforcement learning can improve mathematical reasoning in small language models, but with significant caveats around training instability and length constraints. What is missing is a clean comparison: given a fixed model, fixed compute budget, and fixed evaluation, how much does each training signal—supervised trace imitation versus reward-driven optimization—actually contribute?

We present a controlled study that isolates these contributions. Using Qwen3-1.7B as a common base model with identical LoRA adapter configurations, we compare five training conditions: (1) a **baseline** trained via SFT on 50K question–answer pairs without reasoning traces; (2) **pure distillation**, training via SFT on 1K DeepSeek-R1 reasoning traces from the s1K-1.1 dataset [16]; (3) **pure RL**, applying GRPO with binary answer-correctness rewards directly to the base model; (4) **distillation then RL**, first distilling reasoning traces and then refining with GRPO; and (5) **re-distillation**, training via SFT on 1K RL-verified

reasoning traces from the Open-R1 project. The first four conditions form a 2×2 factorial design—crossing the presence or absence of reasoning traces with the presence or absence of RL—enabling clean attribution of performance to each training signal. The fifth condition tests a key alternative hypothesis: that RL benefits transfer more effectively through higher-quality training data than through direct optimization.

We evaluate all conditions on GSM8K [3], measuring accuracy under greedy decoding across approximately 100 test problems per condition. Our initial hypotheses were: (H1) distillation alone is sufficient for grade-school–level problems; (H2) the combined distillation-then-RL approach outperforms either technique in isolation; (H3) pure RL from the base model struggles at 1.7B scale without a distillation warmup; and (H4) RL-refined models produce shorter, more efficient reasoning traces. Our results confirm H1 and H3 but strikingly *falsify* H2: adding GRPO after distillation degrades performance by 14 percentage points. The re-distillation condition, motivated by this failure, achieves the best overall result (86.0%), demonstrating that the right way to capture RL's benefits at small scale is through data quality, not direct optimization.

Our contributions are: (1) a controlled factorial comparison showing that distillation outperforms direct RL at the 1.7B scale; (2) empirical evidence that naive SFT-then-GRPO causes catastrophic forgetting, with theoretical grounding in the non-decouplability of SFT and RL objectives [26]; and (3) a demonstration that re-distillation from RL-verified traces outperforms both direct distillation and direct RL with identical data budgets.

2 Related Work

Chain-of-thought reasoning. Wei et al. [20] demonstrated that prompting large language models with step-by-step reasoning exemplars dramatically improves performance on arithmetic, commonsense, and symbolic reasoning tasks. Kojima et al. [9] extended this finding by showing that the simple prompt "Let's think step by step" elicits reasoning without any exemplars, establishing that CoT capabilities are latent in sufficiently large models. These results raised a natural question: can smaller models, which do not reliably produce chain-of-thought reasoning through prompting alone, acquire this capability through training?

Knowledge distillation for reasoning. Several approaches have explored training small models to reason by imitating larger ones. Hsieh et al. [6] showed that distilling intermediate rationales—not just final answers—from a teacher model enables a student to outperform standard fine-tuning with substantially less data. Magister et al. [13] distilled chain-of-thought reasoning from PaLM-540B into T5 models, demonstrating that arithmetic and commonsense reasoning transfer across model scales. The Orca line of work [17, 14] progressively refined this approach, moving from imitation of explanation traces to teaching strategy selection. Most recently, Muennighoff et al. [16] demonstrated that fine-tuning on just 1,000 high-quality DeepSeek-R1 traces achieves remarkable performance, suggesting that trace quality matters more than quantity. However, Li et al. [11] identified a critical *learnability gap*: small models often fail to internalize the full reasoning structure of traces from much larger models, instead learning surface-level patterns that do not generalize. All of these works study distillation in isolation; none combine it with reinforcement learning or provide controlled comparisons against RL alternatives.

Reinforcement learning for reasoning. DeepSeek-R1 [5] demonstrated that GRPO can elicit reasoning from a base model without any supervised reasoning data (the "R1-Zero" experiment), and that adding a cold-start SFT phase before RL improves stability and performance. Subsequent work has refined this approach: GRPO-LEAD [21] introduces length-dependent rewards and difficulty-aware reweighting to improve training efficiency, while Liu et al. [12] analyze the role of pretraining quality in RL success and propose Dr. GRPO for improved token-level credit assignment. At small model scale, Dang and Ngo [4] provide a systematic study of GRPO on sub-3B models, identifying learning rate sensitivity and reward hacking as key challenges. AceReason-Nemotron [2] presents evidence that large-scale RL can outperform distillation for small and mid-size models, though their comparison uses different base models and training configurations, making attribution difficult. Our work complements these by providing a controlled comparison—same model, same adapter, same evaluation—that isolates the RL signal from confounding factors.

Efficient fine-tuning for small reasoning models. Low-rank adaptation (LoRA; Hu et al. 7) has become the standard approach for parameter-efficient fine-tuning, but standard LoRA scaling (α/r) causes gradient instability at high ranks. Rank-stabilized LoRA (rsLoRA; Kalajdzievski 8) addresses this by scaling as $\alpha/\sqrt{r}$, enabling stable training at higher ranks where reasoning tasks benefit from increased adapter capacity. Recent work on small reasoning models includes Phi-4-Mini-Reasoning [19], which uses a structured pipeline of distillation, rollout DPO, and RL, and Zhuang et al. [22], who provide a systematic comparison of SFT, distillation, and RL for small reasoning models across multiple model families. Our work differs in its factorial design—rather than comparing complete pipelines, we isolate individual training signals within a single model and configuration.

3 Method

3.1 Experimental Design

Our study compares five training conditions, summarized in Table 1. The first four form a 2×2 factorial design crossing the presence of reasoning traces during SFT with the presence of RL optimization. The fifth condition—re-distillation—tests whether RL benefits can be transferred indirectly through higher-quality training data. All conditions share the same base model, LoRA configuration, optimizer, and evaluation protocol.

Table 1: Experimental conditions. The first four conditions form a factorial design; the fifth tests indirect RL benefit transfer through data quality.

Condition	**Traces**	**RL**	**Training Data**	**What It Isolates**
Baseline	×	×	Orca Math (50K Q&A pairs)	Format-only supervision
SFT-Traces	✓	×	s1K-1.1 (1K R1 traces)	Pure distillation
GRPO-Only	×	✓	GSM8K train (7.5K)	Pure RL from base
SFT→GRPO	✓	✓	s1K-1.1 → GSM8K GRPO	Distillation + RL
Re-Distill	✓	×	Open-R1 (1K verified)	RL benefits via data quality

3.2 Model and Adapter Configuration

We use Qwen3-1.7B [18] as our base model throughout all conditions. At 1.7B parameters, the model sits above the empirical threshold (∼1B) where reasoning capabilities begin to emerge [20], while remaining small enough that training-signal effects are not masked by excess capacity.

We train in full bfloat16 precision without quantization. Although 4-bit NF4 quantization reduces model size from ∼3.4 GB to ∼1.2 GB, QuAILoRA Authors [23] demonstrate measurable accuracy gaps between quantized and full-precision fine-tuning, with the effect proportionally larger for smaller models. Since Qwen3-1.7B fits comfortably on a single L40S (48 GB) even in full precision, we avoid this confound.

We apply LoRA with rank $r = 64$, $\alpha = 64$, and rsLoRA scaling [8] (i.e., the effective scaling factor is $\alpha/\sqrt{r} = 8$ rather than the standard $\alpha/r = 1$). A recent ablation study on Qwen models [25] found that GSM8K performance peaks at $r = 64$ (74.98%) with no improvement at $r = 128$. The adapter targets all attention and MLP projection matrices (q/k/v/o_proj, gate/up/down_proj), totaling 7 modules, with dropout $p = 0.05$.

To ensure compatibility with Qwen3's chat template, we override the tokenizer's EOS token from `<|endoftext|>` to `<|im_end|>`, which the model uses as the end-of-turn marker.

3.3 Training Conditions

Baseline (SFT on Q&A pairs). The baseline condition fine-tunes on 50,000 examples from the Orca Math dataset [15], which contains question–answer pairs without intermediate reasoning steps. This condition isolates the effect of exposure to mathematical problems and answer formatting, without any reasoning

signal. We train for 3 epochs with learning rate 2×10^{-4}, effective batch size 16 (per-device batch 4, gradient accumulation 4), maximum sequence length 4,096, and sequence packing enabled.

SFT-Traces (pure distillation). This condition fine-tunes on the s1K-1.1 dataset [16]: 1,000 problems with complete DeepSeek-R1 reasoning traces wrapped in `<think>...</think>` tags. Training uses the same hyperparameters as the baseline, except the maximum sequence length is increased to 16,384 to accommodate long reasoning traces (which average ∼10K tokens). Three epochs over 1K examples test the few-shot distillation paradigm.

GRPO-Only (pure RL). Starting from the base Qwen3-1.7B (no SFT), we apply GRPO with a binary answer-correctness reward on 10,000 examples from NuminaMath-1.5-RL-Verifiable, following the literature consensus that 7–10K examples represent the sweet spot for RL on math [4]. We use learning rate 1×10^{-6}, 500 training steps, 4 generations per prompt, and maximum completion length 2,048. Generation is performed using vLLM [10] in colocate mode, where the generation engine and the training model share a single GPU via a sleep mechanism that frees VRAM between phases.

SFT→GRPO (distillation then RL). This condition runs the SFT-Traces phase identically, then applies GRPO with a fresh LoRA adapter. The handoff requires merging the SFT adapter into the base weights via `merge_and_unload()`, since TRL does not support stacked LoRA adapters. The GRPO phase uses identical hyperparameters to the GRPO-Only condition, with the merged SFT checkpoint serving as the policy initialization. We note that this merge step is lossy: Shuttleworth et al. [29] show that LoRA produces "intruder dimensions"—high-ranking singular vectors absent from full fine-tuning—that may degrade during weight merging. We discuss this confound in Section 6.

Re-Distill (SFT on RL-verified traces). This condition fine-tunes on 1,000 reasoning traces from the Open-R1 Math dataset (`open-r1/OpenR1-Math-220k`), which contains solutions generated by RL-trained models with automated math verification. We filter for examples where `correctness_math_verify` is `True` and select the first verified-correct generation per problem. The key difference from SFT-Traces is the trace source: Open-R1 traces come from models that underwent RL optimization and were subsequently filtered for mathematical correctness, whereas s1K-1.1 traces are raw DeepSeek-R1 outputs without correctness filtering. Training uses identical hyperparameters to SFT-Traces (same learning rate, epochs, sequence length, batch size), isolating the effect of trace quality from all other variables. This condition tests the *re-distillation* hypothesis [33]: that RL benefits transfer more effectively through higher-quality training data than through direct RL optimization of the small model.

3.4 Reward Functions

All GRPO conditions use two reward functions, combined with configurable weights.

Binary correctness reward. Returns $r = 1$ if the model's extracted answer matches the ground truth, $r = 0$ otherwise. Answer extraction follows a specificity cascade: (1) extract from `\boxed{}` notation with proper brace-depth counting to handle nested expressions (e.g., `\boxed{\frac{1}{2}}`); (2) extract the value after a `####` delimiter (GSM8K format); (3) fall back to the last number in the response. Extracted answers are normalized by stripping currency symbols, percentages, and commas, converting LaTeX fractions to numeric values, and parsing as numbers where possible. Comparison uses `math.isclose` with relative tolerance 10^{-6} and absolute tolerance 10^{-8}, falling back to exact string match.

Format reward. A base model that has never seen reasoning traces cannot produce the `<think>...</think>` format required by the correctness reward's answer extraction. To address this cold-start problem, we add a format reward with partial credit: 0.5 for producing a `<think>...</think>` block, 0.25 for including a `\boxed{}` answer, and 0.25 for producing any number after the thinking block. This gives gradient signal even when answers are incorrect, bootstrapping the model into producing structured output before correctness optimization can take effect. For the GRPO-Only condition, we weight correctness and format equally

(0.5/0.5); for SFT→GRPO, where the model already produces formatted traces, we weight correctness at 1.0 and format at 0.0.

4 Experimental Setup

Benchmark. We evaluate on **GSM8K** [3], which contains 1,319 grade-school math word problems with verifiable numerical answers requiring basic multi-step arithmetic. Due to compute constraints (each model generates long reasoning traces requiring ∼45 minutes per 100 problems on an L40S), we evaluate on the first ∼100 problems of the test set per condition, yielding confidence intervals of approximately ±8 percentage points at 95% confidence. As a reference, Qwen3-1.7B base achieves 75.4% on GSM8K with 4-shot chain-of-thought prompting [18].

Controls and generation. The baseline condition (SFT on Q&A pairs without traces) controls for the effect of mathematical training data exposure, isolating whether reasoning traces—rather than simply more math problems—drive improvement. We use greedy decoding (temperature 0.0) for all accuracy measurements, generating up to 2,048 new tokens per problem with left-padding for batch generation. All experiments are conducted on Modal cloud infrastructure using NVIDIA L40S GPUs (48 GB) for SFT and evaluation, and NVIDIA H100 GPUs (80 GB) for GRPO, which requires additional VRAM for vLLM-based online generation.

5 Results

Table 2: GSM8K accuracy across conditions (∼100 problems per condition, greedy decoding). Qwen3-1.7B base achieves 75.4% with 4-shot chain-of-thought prompting.

Condition	GSM8K Acc.	N	Training Data	Training Signal
Baseline	67.0%	100	50K answers	Answer-only SFT
SFT→GRPO	66.7%	99	1K traces → 7.5K GSM8K	Distillation + RL
GRPO-Only	77.8%	99	7.5K GSM8K	RL from base
SFT-Traces	80.8%	99	1K DeepSeek-R1 traces	Raw distillation
Re-Distill	**86.0%**	93	1K Open-R1 verified traces	RL-verified distillation

5.1 Main Results

The results reveal a clear hierarchy among the five conditions (Table 2). Re-Distill achieves the highest accuracy (86.0%), followed by SFT-Traces (80.8%), GRPO-Only (77.8%), Baseline (67.0%), and SFT→GRPO (66.7%). We discuss each finding in turn.

Distillation is remarkably data-efficient. SFT-Traces (80.8%) outperforms Baseline (67.0%) by 13.8 percentage points despite training on 50× fewer examples (1,000 vs. 50,000). This confirms the central finding of Hsieh et al. [6]: the quality of the training signal—intermediate reasoning rationales versus final answers—matters far more than quantity. Our data efficiency is notable in context: DeepSeek-R1-Distill-Qwen-1.5B achieves 83.9% on the harder MATH-500 benchmark, but uses approximately 800K traces [5]—roughly 800× our dataset.

Interestingly, the Baseline (67.0%) performs *below* the base model's 4-shot capability (75.4%). This is consistent with Rajani et al. [31], who find that SFT modifies mid-layer MLPs aggressively, *replacing* existing capabilities rather than augmenting them. Answer-only SFT appears to teach the model to skip reasoning steps it would otherwise perform in-context, actively degrading its latent chain-of-thought ability.

Re-distillation from RL-verified traces is the best approach. Re-Distill (86.0%) outperforms SFT-Traces (80.8%) by 5.2 points using the same amount of data (1K examples), the same model, and identical hyperparameters. The sole difference is the trace source: math-verified correct solutions from the Open-R1 project (generated by RL-trained models and filtered for correctness) versus raw DeepSeek-R1 traces from s1K-1.1 (unfiltered). This demonstrates that the right way to transfer RL benefits to small models is through *better training data*, not direct RL optimization. The REDI framework [33] established this approach at scale (131K traces); our result extends it to the extreme low-data regime, showing that verification-based filtering matters even at 1K examples.

Direct GRPO after distillation causes catastrophic forgetting. SFT→GRPO (66.7%) performs 14.1 points *worse* than SFT-Traces alone (80.8%), degrading the model to baseline-level performance. This is our most striking result and directly falsifies Hypothesis H2. We analyze the mechanisms behind this degradation in Section 6.

Pure GRPO provides marginal improvement over the base model. GRPO-Only (77.8%) gains only 2.4 points over the base model's 4-shot capability (75.4%), despite 500 training steps on 7,473 GSM8K problems. This confirms Yue et al. [30], who show that RL redistributes sampling probability toward correct paths already present in the base model's distribution but does not create new reasoning capabilities. In contrast, "distillation can genuinely introduce new knowledge"—consistent with our finding that both SFT-Traces (80.8%) and Re-Distill (86.0%) substantially exceed the base model's capability ceiling.

5.2 Training Dynamics

Table 3: Training metrics across SFT conditions (final epoch). All conditions use the same LoRA configuration and optimizer. Lower loss indicates easier learning; higher token accuracy indicates better per-token prediction.

Condition	Final Loss	Token Acc.	Steps	Time (min)
Baseline	0.105	96.2%	669	108
Re-Distill	0.370	86.7%	72	61
SFT-Traces	0.842	75.4%	117	101

The SFT training metrics (Table 3) reveal an instructive pattern. The Baseline achieves the lowest loss (0.105) and highest token accuracy (96.2%)—yet the worst GSM8K performance (67.0%). This is because short answer-only text is easy to memorize but carries no reasoning signal. Conversely, SFT-Traces has the highest loss (0.842) and lowest token accuracy (75.4%), because long reasoning traces from a 671B teacher model are inherently harder for a 1.7B student to imitate—yet this condition achieves 80.8% accuracy. Re-Distill falls between: its Open-R1 traces (average ∼6K tokens) are shorter and easier to learn than s1K traces (∼10K tokens), resulting in lower loss (0.370) and higher token accuracy (86.7%), while achieving the best GSM8K accuracy (86.0%).

GRPO training dynamics tell a different story. For GRPO-Only, 75–85% of training steps had zero within-group reward variance (`frac_reward_zero_std` ≈ 0.80), meaning all generations in a group received identical rewards and GRPO could not compute a meaningful advantage. The model's binary correctness reward averaged near zero across training, with only sporadic steps producing non-trivial gradient signal. For SFT→GRPO, the reward signal was somewhat stronger (mean reward 0.81, reflecting the distilled model's ability to solve GSM8K problems), but `frac_reward_zero_std` remained 0.77—the model was already too accurate on GSM8K for GRPO to find sufficient within-group variance.

5.3 Qualitative Analysis

To assess whether the distilled model learns genuine reasoning rather than surface-level pattern mimicry, we compare outputs from Baseline and SFT-Traces on the same GSM8K problem:

Problem: "A store sells apples for $2 each. If John buys 7 apples and pays with a $20 bill, how much change does he get?"

Baseline response:

> <think></think>
> John buys 7 apples at $2 each, so the total cost of the apples is: 7 apples * $2/apple = $14. John pays with a $20 bill, so the change he gets back is: $20 - $14 = $6.

SFT-Traces response:

> <think> Okay, let me try to figure out this problem. So, John buys 7 apples, each costing $2, and he pays with a $20 bill. We need to find out how much change he gets. First, I know that the price per apple is $2. If he buys 7 apples, the total cost should be 7 multiplied by $2. Let me write that out: 7 * $2 = $14. John paid with a $20 bill. The change he should get back is $20 minus $14 = $6. Let me double-check: 7 apples at $2 each is $14. $20 minus $14 is $6. </think> John gets $\boxed{6}$ dollars in change.

Both models reach the correct answer on this simple problem. However, the reasoning processes are qualitatively different. The Baseline produces an empty <think> block—it learned the format token from the chat template but skips straight to computation. The distilled model restates the problem, identifies relevant quantities, performs the calculation, and *self-verifies* ("Let me double-check"). This self-verification behavior, which the model learned from DeepSeek-R1's reasoning traces, is the mechanism by which distillation improves accuracy on harder problems: when the first computation attempt is incorrect, self-verification triggers a correction that the baseline model cannot perform.

6 Analysis and Discussion

6.1 Hypothesis Evaluation

Table 4 summarizes the status of our four initial hypotheses.

Table 4: Hypothesis evaluation. Two hypotheses are supported, one is falsified, and one is not observed.

Hypothesis	**Result**	**Evidence**
H1: Distillation suffices for easy math	Supported	80.8% on GSM8K from 1K traces
H2: RL refines distilled reasoning	**Falsified**	66.7% $<$ 80.8% ($-$14.1pp)
H3: Pure RL needs scale	Supported	77.8% $\approx$ 75.4% base (+2.4pp)
H4: RL compresses traces	Not observed	GRPO degraded, not compressed

6.2 Why Does SFT→GRPO Degrade?

The 14.1-point degradation from SFT-Traces to SFT→GRPO is our most counterintuitive result. Three independent lines of evidence suggest this is not merely an implementation artifact:

Theoretical non-decouplability. Niu et al. [26] prove on Qwen3-0.6B that SFT and RL objectives are mathematically coupled: when GRPO begins after SFT, "the [SFT] loss increases abruptly and eventually exceeds the value observed for the original base model." Our result is a direct empirical instance of their Theorem 3.1.

Distribution mismatch. Zhang et al. [27] show that stronger SFT checkpoints can *underperform* weaker ones after RL, because the offline SFT data distribution mismatches the on-policy rollout distribution that GRPO uses for advantage estimation. Our SFT model generates reasoning traces very different from what the base model would produce, creating instability during the GRPO phase.

Entropy collapse. Wang et al. [28] demonstrate that standard SFT reduces generation diversity (entropy), leaving RL with a narrowed exploration space. Our GRPO training confirms this: 77% of steps had zero within-group reward variance, meaning the SFT-trained model produced near-identical completions for each prompt—exactly the signature of an entropy-collapsed policy.

LoRA merge artifacts. Our pipeline required merging the SFT LoRA adapter via `merge_and_unload()` before applying a fresh LoRA for GRPO. Shuttleworth et al. [29] show that LoRA produces "intruder dimensions" in the weight matrix's SVD structure that are absent from full fine-tuning. These intruder dimensions may be poorly preserved during merging, causing information loss. A controlled ablation using full fine-tuning for both SFT and GRPO (bypassing the merge step) would isolate this confound.

We note that the SFT→GRPO degradation is not widely reported in prior work because most studies use 7B+ models. At 7B, AceReason-Nemotron [2] shows RL improves distilled models by +14.6% on AIME 2025. The critical insight is **scale-dependent**: SFT-then-RL is complementary at sufficient model capacity, but destructive at 1.7B where the model cannot simultaneously maintain distilled knowledge and accommodate RL-driven weight updates.

6.3 Scale-Dependence of RL for Reasoning

Table 5: RL effectiveness across model scales. Results from this work and published literature. "From base" denotes RL applied to a pretrained model without prior distillation; "from distilled" denotes RL applied after reasoning trace SFT.

Scale	RL from Base	RL from Distilled	Source
0.5B	34.2% GSM8K	54.0% GSM8K	Zhuang et al. [22]
1.7B	**77.8% GSM8K**	**66.7% GSM8K (degraded)**	**This work**
7B	82.2% MATH-500	+14.6% AIME 2025	SimpleRL; Chen et al. [2]
32B	50% AIME 2024	—	DAPO

Table 5 shows a clear pattern: RL effectiveness increases with model scale, and the transition from "RL degrades distillation" to "RL improves distillation" occurs somewhere between 1.7B and 7B parameters. Tan et al. [32] establish a power-law relationship between model size and RL learning efficiency across the Qwen2.5 series, with larger models exhibiting consistently superior efficiency on both compute and data metrics. Our results place the 1.7B scale firmly below the threshold where direct RL optimization becomes beneficial.

6.4 The Re-Distillation Insight

The success of Re-Distill (86.0%) relative to both SFT-Traces (80.8%) and GRPO-Only (77.8%) reveals that the distillation-vs-RL framing is a false dichotomy. The relevant question is not *whether* to use RL, but *how* to transfer its benefits to small models.

Direct RL optimization fails at 1.7B because the model lacks capacity to explore effectively (too few correct answers to create reward variance) and cannot maintain SFT-learned patterns while adapting to the RL objective. Re-distillation sidesteps both problems: a larger model performs the RL exploration, correctness verification filters the results, and the small model receives only the cleaned output via standard SFT. The small model never encounters the instability, forgetting, or engineering complexity of direct RL—yet captures RL's benefits through higher-quality training data.

This result connects to a broader principle: at small scale, **data quality is a more effective lever than optimization algorithm**. The 5.2-point improvement from filtering traces for correctness (SFT-Traces → Re-Distill) exceeds the 2.4-point improvement from 500 steps of GRPO optimization (base → GRPO-Only), and it avoids the 14.1-point degradation risk of direct SFT→GRPO.

7 Limitations

Our study has several limitations that qualify the findings. First, we evaluate on approximately 100 GSM8K problems per condition, yielding confidence intervals of ±8 percentage points. The 5.2-point difference between SFT-Traces and Re-Distill, while suggestive, is not statistically significant at $p < 0.05$; evaluation on the full 1,319-problem test set would strengthen this comparison.

Second, the SFT-Traces and Re-Distill conditions use different trace sources (s1K-1.1 vs. Open-R1), which differ not only in RL verification but also in problem distributions, teacher model identities, and curation methodology. The improvement may partly reflect differences in problem coverage rather than solely trace quality. A controlled comparison using traces from the same problems—one set verified, one not—would isolate the verification effect.

Third, the SFT→GRPO degradation may be partly attributable to the lossy LoRA `merge_and_unload()` step rather than a fundamental SFT-RL conflict. An ablation using full fine-tuning for both phases, or adapter composition methods [29], would isolate this confound.

Fourth, we study a single model family (Qwen3), a single benchmark (GSM8K), and a single reasoning domain (grade-school mathematics). Results may not generalize to other architectures, harder benchmarks (MATH, AIME), or non-mathematical reasoning (code, logic, commonsense).

Finally, all accuracy results use greedy decoding (temperature 0.0). Pass@k evaluation with temperature sampling would reveal whether GRPO improves sampling diversity—even if it does not improve peak capability—as suggested by Yue et al. [30].

8 Future Work

Several directions emerge from our findings.

Isolating the LoRA merge confound. The SFT→GRPO degradation may be partly an artifact of the lossy `merge_and_unload()` step. A controlled ablation using full fine-tuning for both SFT and GRPO phases—eliminating the adapter merge entirely—would isolate the LoRA-specific component from the fundamental SFT-RL objective conflict proved by Niu et al. [26]. Alternatively, adapter composition methods such as DO-Merging [29] or LoRA stacking could preserve adapter structure across training phases.

Entropy-preserving distillation for RL compatibility. If the SFT-then-RL degradation is primarily caused by entropy collapse [28] rather than the merge step, then entropy-preserving SFT methods (CurioSFT) could maintain exploration capacity during distillation, enabling subsequent GRPO to function as intended. The PEAR framework [27] offers another approach: reweighting SFT loss with importance sampling to prepare the distribution for on-policy RL. Testing these methods at 1.7B scale would determine whether the SFT-RL conflict can be mitigated at small scale or whether it is fundamentally capacity-limited.

Optimal trace selection for re-distillation. Our Re-Distill condition uses the first math-verified correct trace per problem. Alternative selection strategies—choosing the shortest correct trace (for reasoning efficiency), the most diverse set of traces (for coverage), or including both correct and incorrect traces with contrastive objectives (following REDI; Xu et al. 33)—could further improve results. The 5.2-point gap between raw and verified traces suggests significant headroom from better data curation.

Scale threshold identification. Our results, combined with the literature (Table 5), suggest a critical model-size threshold between 1.7B and 7B where direct RL transitions from destructive to beneficial after SFT. A systematic study across the Qwen3 family (0.6B, 1.7B, 4B, 8B) with identical experimental design would pinpoint this threshold and determine whether it is a sharp transition or a gradual improvement.

Harder benchmarks and cross-domain generalization. GSM8K measures grade-school arithmetic. Extending evaluation to MATH 500 (competition mathematics), AIME (olympiad-level), and non-mathematical

reasoning (code generation, logical deduction) would test whether the distillation-over-RL advantage generalizes across difficulty levels and domains, or whether RL becomes necessary for problems that exceed the training distribution.

9 Conclusion

For small language models (sub-2B parameters), the path to strong mathematical reasoning runs through data quality, not RL optimization. Our controlled five-condition study on Qwen3-1.7B reveals a clear hierarchy of training signals: RL-verified reasoning traces (86.0%) outperform raw teacher traces (80.8%), which outperform RL from the base model (77.8%), which outperforms answer-only training (67.0%). Most strikingly, adding GRPO after distillation *degrades* performance by 14 percentage points (80.8% $\rightarrow$ 66.7%), a result consistent with recent theoretical work on the non-decouplability of SFT and RL objectives.

The practical recommendation is clear: let larger models do the RL, verify their outputs for correctness, and distill the verified traces into your small model. This *re-distillation* approach captures RL's benefits—higher-quality, more reliable reasoning paths—without its risks: instability, catastrophic forgetting, and engineering complexity. One thousand verified traces and a single SFT run produce a stronger model than weeks of GRPO debugging, at a fraction of the compute cost.

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